Chapter 2 : Integration

your mother's favorite rapper

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1 Chapter 2 : Integration

Problem 1.1. Let $f : X \to \overline{\mathbb{R}}$ and $Y = f^{-1}(\mathbb{R})$. Then f is measurable iff $f^{-1}(\{-\infty\}) \in \mathcal{M}$, $f^{-1}(\{\infty\}) \in \mathcal{M}$, and $f|_Y$ is measurable on Y.

Solution. For the only if direction, $\{\infty\}, \{-\infty\}, \mathbb{R} = Y \in \mathcal{B}_{\overline{\mathbb{R}}}$ making f^{-1} map them into $\mathcal{M}$ as f is measurable and as $\mathcal{M}$ is a sigma algebra we see that $f|_Y : Y \to \overline{\mathbb{R}}$ is $\mathcal{M}_Y$-measurable as $f^{-1}(E) \cap Y \in \mathcal{M}_Y$ for all $E \in \mathcal{B}_{\overline{\mathbb{R}}}$. Now for the if direction, as $\mathcal{B}_{\overline{\mathbb{R}}} = \{E \cup I : E \in \mathcal{B}_{\mathbb{R}}, I \subseteq \{\infty, -\infty\}\}$ we see that for any such $E \cup I \in \mathcal{B}_{\overline{\mathbb{R}}}$ we have $f^{-1}(E \cup I) = f^{-1}(E) \cup f^{-1}(I)$ and as $f^{-1}(E) \cap Y = f^{-1}(E) \in \mathcal{M}_Y \subseteq \mathcal{M}$ from and $f^{-1}(I) \in \mathcal{M}$ so f is measurable on $\mathcal{M}$. □

Problem 1.2. Suppose $f, g : X \to \overline{\mathbb{R}}$ are measurable.

a. fg is measurable (where $0 \cdot (\pm\infty) = 0$).

b. Fix $\alpha \in \overline{\mathbb{R}}$ and define $h(x) = \alpha$ if $f(x) = -g(x) = \pm\infty$ and $h(x) = f(x) + g(x)$ otherwise. Then h is measurable.

Solution.

a. Consider the two maps,

$$X \ni x \longrightarrow (f(x), g(x)) \text{ and } \overline{\mathbb{R}}^2 \ni (x, y) \mapsto xy$$

the first is a measurable map from X (with sigma algebra $\mathcal{M}$) to $R \times R$ (with sigma algebra $\otimes_1^2 \mathcal{B}_{\overline{\mathbb{R}}}$) and the second from this to $\overline{\mathbb{R}}$ (with sigma algebra $\mathcal{B}_{\overline{\mathbb{R}}}$) and as the composition of measurable maps are measurable we are done. Here, the measurability of $(x, y) \to xy$ follows from its continuity (note: the metric on $\overline{\mathbb{R}}$ considered throughout the chapter is usually $d(x, y) := |\arctan(x) - \arctan(y)|$ with $\arctan(\pm\infty) = \pm\pi/2$).

b. Similarly again consider $x \mapsto (f(x), g(x))$ but this time with $(x, y) \to J(x, y)$ where,

$$J(x, y) := \begin{cases} \alpha \text{ if } x = -y = \pm\infty \\ x + y \text{ otherwise} \end{cases}$$

We will use Problem 1.5 here. Let $A = \{(+\infty, -\infty), (-\infty, +\infty)\}$ and $B = \overline{\mathbb{R}}^2 \setminus A$, it is clear that J is continuous on B and hence measurable on it and as it is constant on A it is trivially measurable on it so we are done. As compositions of measurable functions are measurable we are done.

□

Problem 1.3. If $\{f_n\}$ is a sequence of measurable functions on X, then $\{x : \lim f_n(x) \text{ exists}\}$ is a measurable set.

Solution. We know that $g_1 = \limsup f_n$ and $g_2 = -\liminf f_n$ are measurable functions, define

$$h(x) := \begin{cases} 2026 \text{ if } g_1(x) = -g_2(x) = \pm\infty \\ g_1(x) + g_2(x) \text{ otherwise} \end{cases}$$

h is a measurable function by Problem 1.2 and f_n converges iff $g_1 = -g_2$ and we get that

$$\{x : \lim f_n(x) \text{ exists}\} = h^{-1}(\{0, 2026\})$$

which implies that the set infact is measurable as all the sets we performed unions and intersections on are in $\mathcal{M}$. $\square$

Problem 1.4. If $f : X \to \overline{\mathbb{R}}$ and $f^{-1}((r, \infty]) \in \mathcal{M}$ for each $r \in \mathbb{Q}$, then f is measurable.

Solution. It can easily be shown that these sets being in $\mathcal{M}$ implies that $f^{-1}((k, \infty]) \in \mathcal{M}$ for any $k \in \overline{\mathbb{R}}$ easily so we are done. $\square$

Problem 1.5. If $X = A \cup B$ where $A, B \in \mathcal{M}$, a function f on X is measurable iff $f|_A$ is measurable on A and $f|_B$ is measurable on B.

Solution. If f is measurable then measurability on $A, B \in \mathcal{M}$ are trivial. For the other direction, suppsoe $f : X \to Y$ with $\mathcal{N}$ being the sigma algebra on Y, for any $E \in \mathcal{N}$ we have,

$$f^{-1}(E) = (f^{-1}(E) \cap A) \cup (f^{-1}(E) \cap B) = (f|_A^{-1}(E)) \cup (f|_B^{-1}(E)) \in \mathcal{M}$$

as these two sets are in $\mathcal{M}_A, \mathcal{M}_B \subseteq \mathcal{M}$ respectively anyways. $\square$

Problem 1.6. The supremum of an uncountable family of measurable $\mathbb{R}$-valued functions on X can fail to be measurable (unless the σ-algebra $\mathcal{M}$ is very special).

Solution. Suppose $X = \mathbb{R}$ with the sigma algebra being the lebesgue measurable sets, then for a non measurable set $E \subseteq \mathbb{R}$ consider the family $\{\chi_{\{r\}}\}_{r \in E}$, these are all measurable clearly whereas $\sup_{r \in E} \chi_{\{r\}} = \chi_E$ is clearly non measurable unless $\mathcal{M}$. $\square$

Problem 1.7. Suppose that for each $\alpha \in \mathbb{R}$ we are given a set $E_\alpha \in \mathcal{M}$ such that $E_\alpha \subset E_\beta$ whenever $\alpha < \beta$, $\bigcup_{\alpha \in \mathbb{R}} E_\alpha = X$, and $\bigcap_{\alpha \in \mathbb{R}} E_\alpha = \varnothing$. Then there is a measurable function $f : X \to \mathbb{R}$ such that $f(x) \leq \alpha$ on E_α and $f(x) \geq \alpha$ on E_α^c for every α. (Use Exercise 4.)

Solution. We claim that the following function works

$$f(x) := \sup\{r \in \mathbb{Q} : x \notin E_r\}$$

For $x \in E_\alpha$, from the definition we have that $x \in E_\beta$ for all $\mathbb{Q} \ni \beta \geq \alpha$ and thus $f(x) = \sup\{r \in \mathbb{Q} : r < \alpha, x \notin E_r\} \leq \alpha$. Now, for $x \in E_\alpha^c$ its clearly $\geq \alpha$. We claim that, for $r \in \mathbb{Q}$ we have

$$f^{-1}((r, \infty)) \bigcup_{\mathbb{Q} \ni \alpha \geq r} E_\alpha^c$$

If $f(x) > r$, then the set of β for which $x \in E_\beta^c$ (its an open ray unbounded below) must contain r, i.e its of form $(-\infty, v)$ with $v > r$ and we can find a rational $q \in (r, v)$ such that $x \in E_q$ so we have one direction of the equality, the other direction is trivial and we are done as the set on the right is in $\mathcal{M}$ being a countable union of sets from $\mathcal{M}$ which is sufficient to prove measurablity via Problem 2.4 $\square$

Problem 1.8. If $f : \mathbb{R} \to \mathbb{R}$ is monotone, then f is Borel measurable.

Solution. trivial. □

Problem 1.9. Let $f : [0,1] \to [0,1]$ be the Cantor function (§1.5), and let $g(x) = f(x) + x$.

a. g is a bijection from $[0,1]$ to $[0,2]$, and $h = g^{-1}$ is continuous from $[0,2]$ to $[0,1]$.

b. If C is the Cantor set, $m(g(C)) = 1$.

c. By Exercise 29 of Chapter 1, $g(C)$ contains a Lebesgue nonmeasurable set A. Let $B = g^{-1}(A)$. Then B is Lebesgue measurable but not Borel.

d. There exist a Lebesgue measurable function F and a continuous function G on $\mathbb{R}$ such that $F \circ G$ is not Lebesgue measurable.

Solution.

a. It is known that the cantor function is continuous on $[0,1]$ as well as being increasing. This makes $g(x) = f(x) + x$ strictly increasing and continuous, with this we can say using the intermediate value theorem that g is is a continuous bijection from $[0,1]$ to $[0,2]$ and $h = g^-1$ is also continuous.

b. As the cantor set is borel and h is continuous, it must be borel measurable. Thus, $h^{-1}(C) = g(C)$ is borel (as well as being lebesgue measurable which is weaker). We can write $m(g([0,1])) = m(g(C)) + m(g([0,1] \setminus C))$ as the sets here are disjoint (the second one is clearly measurable). From the previous parts we see that $m(g(C)) = 2 - m(g([0,1] \setminus C))$. This set is the disjoint union of the middle third intervals removed in the process of constructing the cantor set, on which, by defintion the cantor function is constant. These removed open intervals have measures summing to 1 and as g is a translation on these we see that $m(g([0,1] \setminus C)) = 1$ showing that $m(g(C)) = 1$.

c. As the lebesge measure is complete we see that B being the subset of the cantor set which has measure zero, must be lebesgue measurable too. Now, if B was borel then so would be $h^-1(B) = A$ as g^{-1} is continuous, so it is not borel.

d. Consider $\chi_B \circ h$, we see that $(\chi_B \circ h)^{-1}(\{1\}) = h^{-1}(\chi_B^{-1}(\{1\})) = h^{-1}(B) = A$ which is not lebesgue measurable while we had h be continuous (a homomorphism even) and χ_B lebesgue measurable so we are done as $\{1\}$ is borel.

□

Problem 1.10. Prove that the following implications are valid iff the measure μ is complete:

a. If f is measurable and $f = g$ μ-a.e. then g is measurable.

b. If f_n is measurable for $n \in \mathbb{N}$ and $f_n \longrightarrow f$ μ-a.e. then f is measurable.

Solution.

a. First assume that μ is complete and that f and g disagree on some $N \in \mathcal{M}$ with $\mu(N) = 0$, then for any measurable E in the range we must have that

$$g^{-1}(E) = (g^{-1}(E) \cap N^c) \cup (g^{-1}(E) \cap N) = (g|_{N^c}^{-1}(E)) \cup (g^{-1}(E) \cap N)$$
$$= \underbrace{(f|_{N^c(\in\mathcal{M})}^{-1}(E))}_{\in\mathcal{M}} \cup \underbrace{(g^{-1}(E) \cap N)}_{\subseteq N \implies \in\mathcal{M}}$$

and thus this set must be in $\mathcal{M}$ itself. For the other direction, suppose $\mu(N) = 0$ for some $N \in \mathcal{M}$ and $K \subseteq N$. Let, $f = \chi_N$ and $g = \chi_K$, we see that $f = g$ on N^c and thus $f = g$ μ-a.e. which makes g measurable and thus $g^{-1}(\{1\}) = K$ must also be measurable making μ complete.

b. For one direction assume that μ is complete. By Problem 2.3 we know that the set where we have convergence, is measurable so let it be E. Further, a.e. convergence implies that $\mu(X \setminus E) = 0$. Let $g = \lim \chi_E f_n(x)$, we see that g is measurable on E as well as on E^c and is thus measurable on X, being the limit (it always exists) of measurable functions. Suppose, N is the set where $f_n \not\longrightarrow f$, by our hypothesis this is of measure zero. For $x \in N^c$ however $f_n(x) \longrightarrow f(x)$ and thus $f(x) = g(x)$ so $f(x) \neq g(x)$ can only happen on N implying that $f = g$ a.e after which part a proves measurablity of f and we are done. For the other direction, suppose E is a set of measure zero and $K \subseteq E$. Let, $f = \chi_K$ and $f_n = 0$, with this we see that $f_n \to f$ on E^c which shows that $f_n \to f$ a.e. and thus f is measurable so we are done.

□

Problem 1.11. Suppose that f is a function on $\mathbb{R} \times \mathbb{R}^k$ such that $f(x, \cdot)$ is Borel measurable for each $x \in \mathbb{R}$ and $f(\cdot, y)$ is continuous for each $y \in \mathbb{R}^k$. For $n \in \mathbb{N}$, define f_n as follows. For $i \in \mathbb{Z}$ let $a_i = i/n$, and for $a_i \leq x \leq a_{i+1}$ let

$$f_n(x, y) = \frac{f(a_{i+1}, y)(x - a_i) - f(a_i, y)(x - a_{i+1})}{a_{i+1} - a_i}$$

Then f_n is Borel measurable on $\mathbb{R}\times\mathbb{R}^k$ and $f_n \to f$ pointwise; hence f is Borel measurable on $\mathbb{R} \times \mathbb{R}^k$. Conclude by induction that every function on $\mathbb{R}^n$ that is continuous in each variable separately is Borel measurable.

Solution. We can write $\mathbb{R} \times \mathbb{R}^k$ as a countable union of the borel sets $\{[a_i, a_{i+1}] \times \mathbb{R}^k\}_{i\in\mathbb{Z}}$ and proving borel measurablity of f_n on these suffices to prove borel measurablity globally. It being borel on $[a_i, a_{i+1}] \times \mathbb{R}^k$ follows from these maps being clearly borel

$$(x, y) \mapsto (x - a_i), (x, y) \mapsto (x - a_{i+1}), (x, y) \mapsto f(a_i, y), (x, y) \mapsto f(a_{i+1}, y)$$

as the borel measurability is preserved under products and sums. To prove pointwise convergence just notice that for $(x, y) \in [a_i, a_{i+1}] \times \mathbb{R}^k$, $f_n(x, y)$ is between $f(a_i, y)$ and $f(a_{i+1}, y)$ which converge to $f(x, y)$ as $n \to \infty$ by continuity of $f(\cdot, y)$. This shows that if $f(\cdot, y)$ is continuous and $f(x, \cdot)$ borel, for all $(x, y) \in \mathbb{R}\times\mathbb{R}^k$ then f is borel. Now, suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is continuous in each variable, suppose that the proposition holds for $n - 1$, then for all $x \in \mathbb{R}$, $f(x, \cdot) : y \in \mathbb{R}^{n-1} \mapsto f(x, y)$ is borel because this is continuous in each variable as well and for all $y \in \mathbb{R}^{n-1}$, $f(\cdot, y) : x \in \mathbb{R} \mapsto f(x, y)$ is continuous too,

which implies that f is borel. We can induct on n from the base case of $n = 1$ where the proposition is true due to continuity in each variable coinciding with continuity (which is stronger than borel measurability), hence proving the proposition. $\square$

Problem 1.12. Prove Proposition 2.20. (See Proposition 0.20, where a special case is proved.)

Solution. If $f^{-1}(\{\infty\})$ is not null then the integral will trivially be infinite and thus it must be a null set. Now, consider the sets $H_n = f^{-1}(1/n, \infty]$ and we find that $\{x : f(x) > 0\} = \cup H_n$. Here, we see that $f \geq n^{-1}\chi_{H_n}$ for all $n \geq 1$ and thus $\int f \geq n^{-1}\mu(H_n)$ forces all the H_n to have finite measure. $\square$

Problem 1.13. Suppose $\{f_n\} \subset L^+$, $f_n \to f$ pointwise, and $\int f = \lim \int f_n < \infty$. Then $\int_E f = \lim \int_E f_n$ for all $E \in \mathcal{M}$. However, this need not be true if $\int f = \lim \int f_n = \infty$.

Solution. For any $E \in \mathcal{M}$ we can see that $f_n\chi_E \longrightarrow f\chi_E$ pointwise as well and thus, by fatou's lemma and the additivity of the integral we have,

$$\int f\chi_E \leq \liminf \int f_n\chi_E \text{ and } \int f\chi_{X\setminus E} \leq \liminf \int f_n\chi_{X\setminus E}$$

$$\implies \int f = \int f\chi_E + \int f\chi_{X\setminus E} \leq \liminf \int f_n\chi E + \liminf \int f\chi_{X\setminus E}$$

Using the fact that the limit infimum is subadditive we see that the above is

$$\leq \liminf \left(\int f_n\chi_E + f_n\chi_{X\setminus E} \right) = \liminf \int f_n = \int f$$

as all the terms involved in this string of inequalities are finite (this follows from the fact that $\int f < \infty$ and that $\int f_n$ is close to $\int f$ eventually) we must have that,

$$\int f\chi_E = \liminf \int f_n\chi_E$$

Assume wlog $\int f_n$ is always finite, then by the superadditivity of the limit supremum and the equality we developed earlier, we have

$$\limsup \int f_n\chi_E = \limsup \left(\int f_n - \int f_n\chi_{E^c} \right) \leq \limsup \int f_n + \limsup \left(-\int f_n\chi_{E^c} \right)$$

$$= \limsup \int f_n - \liminf \int f_n\chi_{E^c} = \int f - \int f\chi_{E^c} = \int f\chi_E$$

$$\limsup \int f_n\chi_E \leq \int f\chi_E = \liminf f_n\chi_E \implies \lim \int f_n\chi_E = \int f\chi_E$$

For a counterexample we can consider a function on $\mathbb{R}$ given by $f_n = n\chi_{(0,1/n)} + \chi_{[1,\infty)}$ which converges pointwise to $\chi_{[1,\infty)}$ and $\int f = \lim \int f_n = \infty$ but if we focus on $(0, 1)$ then $\lim f_n\chi_{(0,1)} = 1$ whereas $\int f\chi_{(0,1)} = 0$. Here the area escaped to infinity in the unit open interval and hence the contradiction. $\square$

Problem 1.14. If $f \in L^+$, let $\lambda(E) = \int_E f\, d\mu$ for $E \in \mathcal{M}$. Then λ is a measure on $\mathcal{M}$, and for any $g \in L^+$, $\int g\, d\lambda = \int gf\, d\mu$. (First suppose that g is simple.)

Solution. For any disjoint collection $\{E_n\}_{n\in\mathbb{N}} \subseteq \mathcal{M}$ with union E then by the sigma additivity of the integral we have,

$$\lambda(E) = \int f\chi_E d\mu = \int \sum_{n\in\mathbb{N}} f\chi_{E_n} d\mu = \sum_{n\in\mathbb{N}} \int f\chi_{E_n} d\mu = \sum_{n\in\mathbb{N}} \lambda(E_n)$$

Now, suppose $\phi = \sum a_j \chi_{E_j}$ is a simple function, then we see that

$$\int \phi d\lambda = \sum a_j \lambda(E_j) = \sum a_j \int f\chi_{E_j} d\mu = \int \sum a_j f\chi_{E_j} d\mu = \int f\phi d\mu$$

Now, if $g \in L^+$, we can find a sequence of simple functions $\{\phi_n\}$ that increase to g, for these we also see that $f\phi_n$ iincreases to fg and by the monotone convergence theorem and the fact that λ and μ are defined on the same sigma algebra $\mathcal{M}$ we have that,

$$\int fg d\mu = \int \lim f\phi_n d\mu \longleftarrow \int f\phi_n d\mu = \int \phi_n \lambda \longrightarrow \int \lim \phi_n \lambda = \int g\lambda$$

where we used the the monotone convergence theorem on both sides. □

Problem 1.15. If $\{f_n\} \subset L^+$, f_n decreases pointwise to f, and $\int f_1 < \infty$, then $\int f = \lim \int f_n$.

Solution. Clearly we have $\int f \leq \lim \int f_n < \infty$. For the other direction, consider the measurable set $E = \{f > 0\}$. Fix $\alpha > 1$ and define $F_n := \{x \in E : f_n(x) \leq \alpha f(x)\}$, then $\{F_n\}$ is an increasing sequence of measurable sets and $\cup F_n = E$. From the definiton we see that $\int_{F_n} f_n \leq \alpha \int_{F_n} f$ and $\int_E f_n - \int_{F_n} f_n = \int_{E\setminus F_n} f_n \leq \int_{E\setminus F_n} f_1$. As $A \mapsto \int_A f_1$ is a measure on $\mathcal{M}$ and $E \setminus F_n$ is a decreasing sequence with $\int_{E\setminus F_1} f_1 \leq \int f_1 < \infty$ we see that

$$\lim_{n\to\infty} \int_{F\setminus F_n} f_1 = \int_{\cap E\setminus F_n} f_1 = \int_{E\setminus E} f_1 = 0$$

this implies that $\lim \int_{F_n} f_n = \lim \int_E f_n$. As $A \mapsto \int_A f$ is also a measure, we have

$$\lim \int_{F_n} f_n = \lim \int_E f_n \implies \lim \int_E f_n \leq \lim \alpha \int_{F_n} f = \alpha \lim \int_{F_n} f = \alpha \int_{\cup F_n} f = \int_E f$$

As this is true for any $\alpha > 1$ we have $\lim \int_E f_n \leq \int_E f$. Now, fix $\alpha > 0$ and consider E^c on which f is zero and thus f_n decreases to 0 here. Let $G_n := \{x \in E^c : f_n(x) \leq \alpha f_1(x)\}$ see that $\{G_n\}$ is an increasing sequence of measurable sets with $\cup G_n = E^c$. It can also be shown that $\lim \int_{G_n} f_n = \lim \int_{E^c} f_n$ using similar arguements as we did when working over E and from the definition we see that $\int_{G_n} f_n \leq \alpha \int_{G_n} f_1 \leq \alpha \int f_1$ and thus, $\lim \int_{E^c} f_n = 0 = \int_{E^c} f$ as $\alpha > 0$ was arbitrary. Combining these two deductions, namely $\lim \int_E f_n \leq \int_E f$ and $\lim \int_{E^c} f_n = 0 = \int_{E^c} f$ we see that $\lim \int f_n \leq \int f$ as the integral is additive. Having shown both directions of the inequality we must have $\int f = \lim \int f_n$. □

Problem 1.16. If $f \in L^+$ and $\int f < \infty$, for every $\epsilon > 0$ there exists $E \in \mathcal{M}$ such that $\mu(E) < \infty$ and $\int_E f > (\int f) - \epsilon$.

Solution. From Problem 2.12 we see that the set $E = \{x : f(x) > 0\}$ is sigma finite so we write it as a disjoint(standard technique) union of countably many sets of finite measure as $E = \cup F_n$, now from the additivity of the integral we have,

$$\int f = \int f\chi_E + \underbrace{\int f\chi_{E^c}}_{=0} = \int_E f = \sum_{n\in\mathbb{N}} \int_{F_n} f = \lim_n \sum_{k=1}^{n} \int_{F_n} f = \lim_n \int_{\cup_{k\leq n} F_k} f$$

Now, for any $\epsilon > 0$ we can find a N such that $\int f < \epsilon + \int_{\cup_{k\leq N} F_k} f$ and as $\mu(\cup_{k\leq N} F_k) = \sum_{k\leq N} \mu(F_k) < \infty$ we are done. □

Problem 1.17. Assume Fatou's lemma and deduce the monotone convergence theorem from it.

Solution. Fatou's lemma implies that $\lim \int f_n = \liminf \int f_n \geq \int \liminf f_n = \int f$ while the other direction of the inequality is trivial. □

Problem 1.18. Fatou's lemma remains valid if the hypothesis that $f_n \in L^+$ is replaced by the hypothesis that f_n is measurable and $f_n \geq -g$ where $g \in L^+ \cap L^1$. What is the analogue of Fatou's lemma for nonpositive functions?

Solution. Clearly $f_n + g \in L^+$ so we can apply fatou's lemma on it and get

$$\int g + \liminf \int f_n = \liminf \left(\int f_n + \int g \right) = \liminf \int g + f_n$$

$$\geq \int \liminf(f_n + g) = \int g + \liminf f = \int g + \int \liminf f_n$$

and now we can just cancel out the $\int g$ terms. For nonpositive functions f ($-f \in L^+$) we will have that

$$\limsup \int f_n \leq \int \limsup f_n$$

□

Problem 1.19. Suppose $\{f_n\} \subset L^1(\mu)$ and $f_n \to f$ uniformly.

a. If $\mu(X) < \infty$, then $f \in L^1(\mu)$ and $\int f_n \to \int f$.

b. If $\mu(X) = \infty$, the conclusions of (a) can fail. (Find examples on $\mathbb{R}$ with Lebesgue measure.)

Solution.

a. This is trivial.

b. consider $f_n = n^{-1}\chi_{[0,n^2]}$ on $\mathbb{R}$ with respect to the lebesgue measure.

□

Problem 1.20. (A generalized Dominated Convergence Theorem) If $f_n, g_n, f, g \in L^1$, $f_n \to f$ and $g_n \to g$ a.e., $|f_n| \leq g_n$, and $\int g_n \to \int g$, then $\int f_n \to \int f$. (Rework the proof of the dominated convergence theorem.)

Solution. We see that $g_n+f_n, g_n-f_n \in L^1$ and also as $f_n \to f, g_n \to g$ a.e. we can restrict the integrals to a set E where convergence is global such that $\int = \int_E$ (as operators) and we work inside E now without mention. By fatou's lemma

$$\int g + \liminf \int f_n = \liminf \int f_n + \lim \int g_n = \liminf \left(\int f_n + \int g_n \right) = \liminf \int f_n + g_n$$

$$\geq \int \liminf (g_n + f_n) \geq \int \liminf g + \liminf f_n = \int f + g = \int f + \int g$$

cancelling out the $\int g$ terms we get $\liminf \int f_n \geq \int f$ and doing the same with $g_n - f_n$ we get $-\limsup \int f_n = \liminf \int -f_n \geq \int -f = -\int f$ which implies that $\int f \geq \limsup \int f$ and thus finally we have $\limsup \int f_n \leq \int f \leq \liminf \int f_n \implies \lim \int f_n = \int f$. □

Problem 1.21. Suppose $f_n, f \in L^1$ and $f_n \to f$ a.e. Then $\int |f_n - f| \to 0$ iff $\int |f_n| \to \int |f|$. (Use Exercise 20.)

Solution. For the only if direction we can use generalised dominated convergence (Problem 2.20) on $|f_n - f| \to 0$ a.e. where it is dominated by $|f_n| + |f| \to 2|f|$ a.e. as $\int |f_n| + |f| = \int |f| + \int |f_n| \to 2 \int |f| = \int 2|f|$ and for the if direction we can consider $|f_n| \leq |f| + |f_n - f|$ and proceed similarly. □

Problem 1.22. Let μ be counting measure on $\mathbb{N}$. Interpret Fatou's lemma and the monotone and dominated convergence theorems as statements about infinite series.

Problem 1.23. Given a bounded function $f : [a, b] \to \mathbb{R}$, let

$$H(x) = \lim_{\delta \to 0} \sup_{|y-x| \leq \delta} f(y), \quad h(x) = \lim_{\delta \to 0} \inf_{|y-x| \leq \delta} f(y).$$

Prove Theorem 2.28b by establishing the following lemmas:

a. $H(x) = h(x)$ iff f is continuous at x.

b. In the notation of the proof of Theorem 2.28a, $H = G$ a.e. and $h = g$ a.e. Hence H and h are Lebesgue measurable, and $\int_{[a,b]} H \, dm = \overline{I}_a^b(f)$ and $\int_{[a,b]} h \, dm = \underline{I}_a^b(f)$.

Solution.

a. This is trivial.

b. If it was continuous outside a null set then we can show riemann integrablity by considering partitions of form $P = \{a_1 < b_1 < \ldots < a_n < b_n\} \cup P_\delta$ where, by our assumptions $m(E = \cup(a_i, b_i)) = \epsilon$ is small such that f is continuous on the compact set $[a, b] \setminus E$ and P_δ is of small enough mesh such as to make $U(f|_{[a,b]\setminus E}, P_\delta) - L(f|_{[a,b]\setminus E}) < \delta$. Now, as f is bounded there is some M such that $|f| \leq M$ then

$$U(f, P) - L(f, P) \leq 2M\epsilon + \delta$$

which can be made arbitrarily small so f must be integrable. Now we will prove the converse. If f is integrable then there exists an increasing sequence of finite partitions P_k such that $\|P_k\| \to 0$ and $\lim \int G_{P_k} dm = \int f = \lim \int g_{P_k} dm$. It was already shown in the book that $g = G$ a.e. when f is riemann integrable so if we

show that $H = G$ a.e. and $h = g$ a.e. then we would be done using part (a) as this implies $H = h$ a.e. which is equivalent to continuity a.e. by part (a). We will show that $H = G$ a.e. and the rest will follow similarly. Let K be the set of all endpoints of some interval in P_k for some k, this is a null set and we will show that $H = G$ outside it. Say $x \in [a, b] \setminus K$ is any point then we first show that $H \geq G$. First define $H_\delta(x) := \sup_{|x-y|\leq\delta} f(y)$ then we see that $H = \lim_{\delta\to 0} H_\delta$ by definition and $H_\delta \leq H_{\delta'}$ for all $0 \leq \delta \leq \delta'$. Now, for any $\delta > 0$ we can find some k such that $x \in (a_k, b_k) \subseteq (x - \delta, x + \delta)$ where (a_k, b_k) is an interval in P_k as $\|P_k\| \to 0$. Then we must have that $G_{P_k}(x) \leq H_\delta(x)$ and as G_{P_k} is decreasing we have that $G(x) \leq H_\delta(x)$ and then taking the limit as $\delta \searrow 0$ we have $G(x) \leq H(x)$. To show the other direcrion of the equality we notice that for any P_k with the interval (a_k, b_k) containing x within it we can find some $\delta > 0$ such that $(x - \delta, x + \delta) \subseteq (a_k, b_k)$ and this implies that $G_{P_k}(x) \geq H_\delta(x)$ and as $H_\delta(x)$ is decreasing in delta, taking the limit we see that $G_{P_k}(x) \geq H(x)$ and again takinng the limit as $k \to \infty$ we see that $G \geq H$ so we are done.

□

Problem 1.24. Let $(X, \mathcal{M}, \mu)$ be a measure space with $\mu(X) < \infty$, and let $(X, \overline{\mathcal{M}}, \overline{\mu})$ be its completion. Suppose $f : X \to \mathbb{R}$ is bounded. Then f is $\overline{\mathcal{M}}$-measurable (and hence in $L^1(\overline{\mu})$) iff there exist sequences $\{\phi_n\}$ and $\{\psi_n\}$ of $\mathcal{M}$-measurable simple functions such that $\phi_n \leq f \leq \psi_n$ and $\int(\psi_n - \phi_n)\, d\mu < n^{-1}$. In this case, $\lim \int \phi_n\, d\mu = \lim \int \psi_n\, d\mu = \int f\, d\overline{\mu}$.

Solution. To show the if direction we can consider the functions $g := \sup_n \phi_n$ and $h := \inf_n \psi_n$then these are $\mathcal{M}$-measurable and we have that $\int g - h < 1/n$ for all n and thus it is zero which makes $h = g$ a.e. and thus $f = g = h$ μ-a.e. Now, If h, g are $\mathcal{M}$ measurable then they are clearly $\overline{\mathcal{M}}$ measurable as well and as $\mathcal{M} \subseteq \overline{\mathcal{M}}$ we see that $\overline{\mu}$-a.e. too so we must have that $h = f = g$ $\overline{\mu}$-a.e as well. which implies that f is $\overline{\mathcal{M}}$ measurable. For the converse, we first deal with the case where $f \geq 0$ consider the following simple functions which are $\overline{\mathcal{M}}$ measurable,

$$\phi_n^* := \sum_{k=0}^{2^{2n}-1} \frac{k}{2^n} \cdot \chi_{f^{-1}([k2^{-n},(k+1)2^{-n}))}$$

$$\psi_n^* := \sum_{k=0}^{2^{2n}-1} \frac{k+1}{2^n} \cdot \chi_{f^{-1}([k2^{-n},(k+1)2^{-n}))}$$

Then, for all large enough n we have that $\psi_n^* \geq f \geq \phi_n^*$ and $0 \leq f - \phi_n^*, \psi_n^* - f \leq 2^{-n}$. As f is bounded we can find some constant M with $f \leq M$. For brevity write

$$\phi_n^* = \sum_{k=1}^{m} a_k \chi_{E_k \cup F_k} \text{ where } E_i \cup F_i \text{ are disjoint } E_i \in \mathcal{M}, F_i \subseteq N_i \in \mathcal{M} \text{ with } \mu(N_i) = 0$$

Then define, $\phi_n := \sum_{k=1}^{m} a_k \chi_{E_k}$ and we have $\phi_n \leq \phi_n^*$ with equality $\overline{\mu}$ a.e. Again writing ψ_n^* in the exact same notation with $N = \cup N_i \in \mathcal{M}$ we see that

$$\psi_n := \chi_{N^c}\psi_n^* + (M+1)\chi_N = (M+1)\chi_N + \sum_{k=1}^{m} a_k \chi_{E_m \cap N^c}$$

is such that $\psi_n \geq \psi_n^*$ with equality with equality $\overline{\mu}$ a.e. (We use $M+1$ instead of M to guarantee the inequality, we need to do this due to how $\psi_n^* \leq f + 2^{-n} \leq M+1$ for large enough n). We now have $\mathcal{M}$ measurable simple functions ψ_n, ϕ_n such that $\psi_n \geq f \geq \phi_n$ and also $\psi_n = \psi_n^*$ and $\phi_n = \phi_n^*$ $\overline{\mu}$-a.e. (both together), from this we can say that $\psi_n - f, f - \phi_n \leq 2^{-n}$ together $\overline{\mu}$-a.e. implying $\psi_n - \phi_n \leq 2^{-n+1}$ $\overline{\mu}$-a.e. and thus,

$$\int \psi_n - \phi_n d\overline{\mu} = \int \psi_n - \phi_n d\mu \leq \int 2^{-n+1} d\mu = \frac{\mu(X)}{2^{-n+1}} \longrightarrow 0$$

and we can start from a larger index to have $\int \psi_n - \phi_n < n^{-1}$. In the previous line we assumed the following: if g is some $\mathcal{M}$ measurable function then $\int g d\mu = \int g d\overline{\mu}$, this is easy to prove and we omit the proof. We also see that $\phi_n \leq f \leq \phi_n + 2^{-n}$ $\overline{\mu}$-a.e. for large enough n and thus for large n we have $\int \phi_n d\mu \leq \int f d\overline{\mu} \leq \mu(X) \cdot 2^{-n} + \int \phi_n d\mu$ using the assumption from earlier along the way which implies that $\lim \int \phi_n d\mu = \int f d\overline{\mu}$. The same can be proved for ψ_n similarly so we are done with the proof for $f \geq 0$. Now, for general such f we first split it into its measurable (and bounded in this case positive and negative parts, in usual notation $f = f_+ - f_-$. Let ϕ_n^+, ψ_n^+ be as in the previous case, for f_+ and ϕ_n^-, ψ_n^- be those for f_-, then we see that $\phi_n^+ - \psi_n^- \leq f_+ - f_- \leq \psi_n^+ - \phi_n^-$. As $|\phi_n^+ - \psi_n^- - (f_+ - f_-)| \leq |\phi_n^+ - f_+| + |\psi_n^- - f_-| \leq 2^{-n} + 2^{-n}$ $\overline{\mu}$-a.e for large enough n implying

$$\phi_n^+ - \psi_n^- \leq f_+ - f_- \leq \phi_n^+ - \psi_n^- + 2^{-n+1}$$

and we can get the corresponding bounds for $\psi_n^+ - \phi_n^-$ as well and finish like in the previous case using the $\mathcal{M}$ measurable simple functions functions $\zeta_n := \phi_n^+ - \psi_n^-$ and $\gamma_n := \psi_n^+ - \phi_n^-$ with $\zeta_n \leq f \leq \gamma_n$ and $\gamma_n - \zeta_n \leq 2^{-n+2}$ holding $\overline{\mu}$-a.e.

□

Problem 1.25. Let $f(x) = x^{-1/2}$ if $0 < x < 1, f(x) = 0$ otherwise. Let $\{r_n\}_1^\infty$ be an enumeration of the rationals, and set $g(x) = \sum_1^\infty 2^{-n} f(x - r_n)$.

a. $g \in L^1(m)$, and in particular $g < \infty$ a.e.

b. g is discontinuous at every point and unbounded on every interval, and it remains so after any modification on a Lebesgue null set.

c. $g^2 < \infty$ a.e., but g^2 is not integrable on any interval.

Solution.

a. It follows from the monotone convergence theorem and the fact that the lebesgue integral and riemann integral agree that $\int g dm = \sum_{n=1}^\infty \int 2^{-n} f(x - r_n) dm = \sum_{n=1}^\infty 2^{-n} \cdot \int_0^1 x^{-1/2} dx = 2$ implying $g \in L^1(m)$ which again forces $\{g \geq \infty\}$ to be null.

b. given any real y, if $r_n < y < 1 + r_n$ then we see that $g(y) > 2^{-n}(y - r_n)^{-2}$. Now, given any $x \in \mathbb{R}$ and small enough $\delta > 0$ we can find some r_n such that $x - \delta < r_n < x < x + \delta < r_n + 1$ and thus, for all small enough $\epsilon > 0$ with $r_n < r_n + \epsilon < x$ we see that $g(r_n + \epsilon) > 2^{-n}\epsilon^{-2}$ which can be made arbitrarily large, this shows that g is unbounded in any open neighbourhood around x from whcih the claim follows.

c. As $g < \infty$ a.e. we clearly have $g^2 < \infty$ a.e. as well. With the same setup as the previous part, we see that $g(y)^2 > 4^{-n}(y - r_n)^{-1}$ and thus

$$\int_{x-\delta}^{x+\delta} g^2 \mathrm{d}m \geq \int_{r_n}^{x} \frac{1}{2^n(y - r_n)} \mathrm{d}m(y) = \infty$$

thus, its not integrable on any interval of form $(x - \delta, x + \delta)$ proving the claim after noticing how all intervals have an interval of this form.

□

Problem 1.26. If $f \in L^1(m)$ and $F(x) = \int_{-\infty}^{x} f(t)\, dt$, then F is continuous on $\mathbb{R}$.

Solution. First we prove right continuity using the sequential criterion. Say $x_n \searrow x$, then as $A \mapsto \int \chi_A |f|$ is a finite measure on $\mathcal{L}$ we see that

$$\lim |F(x_n) - F(x)| = \lim \left| \int_{[x,x_n]} f \mathrm{d}m \right| \leq \lim \int_{[x,x_n]} |f| \mathrm{d}m = \int_{\cap[x,x_n]} |f| dm = 0$$

as $\cap[x, x_n] = \{x\}$ is a lebesge null set and we follow the convention of $0 \cdot \infty = 0$. Right continuity is proven similarly and hence we have continuity. □

Problem 1.27. Let $f_n(x) = ae^{-nax} - be^{-nbx}$ where $0 < a < b$.

a. $\sum_1^\infty \int_0^\infty |f_n(x)|\, dx = \infty$.

b. $\sum_1^\infty \int_0^\infty f_n(x)\, dx = 0$.

c. $\sum_1^\infty f_n \in L^1([0, \infty), m)$, and $\int_0^\infty \sum_1^\infty f_n(x)\, dx = \log(b/a)$.

Solution. This is just a bunch of calculations. However it furnishes an example to how how $\int \sum |f_n| \notin L^1$ might result in $\int \sum f_n \neq \sum \int f_n$ even whe both are finite and $\sum f_n \in L^1$ □

Problem 1.28. Compute the following limits and justify the calculations:

a. $\lim_{n\to\infty} \int_0^\infty (1 + (x/n))^{-n} \sin(x/n)\, dx$.

b. $\lim_{n\to\infty} \int_0^1 (1 + nx^2)(1 + x^2)^{-n}\, dx$.

c. $\lim_{n\to\infty} \int_0^\infty n \sin(x/n)[x(1 + x^2)]^{-1}\, dx$.

d. $\lim_{n\to\infty} \int_a^\infty n(1 + n^2x^2)^{-1}\, dx$. (The answer depends on whether $a > 0$, $a = 0$, or $a < 0$. How does this accord with the various convergence theorems?)

Solution.

(a) Let, $f_n(x) = (1 + x/n)^{-n} \sin(x/n)$ then it converges $e^{-x} \cdot \sin(0) = 0$ pointwise and it can be checked that $|f_n| \leq (1 + x/2)^{-2}$ for all $n \geq 2$ and as this is L^1, it allows us to conclude that the limit is 0 by DCT.

(b) Let $f_n(x) = (1 + nx^2)/(1 + x^2)^n$ then it converges to $\chi_{\{1\}}$ pointwise and as $|f_n| \leq 1$ in the interval $[0, 1]$ we can apply DCT to conclude that the limit is 0

(c) In a similar fashion we see that the integrands are dominated by $(1+x^2)^{-1}$ and converge pointwise to it as well after which we can conclude with DCT that the limit is $\pi/2$ using basic calculus.

(d) If $a > 0$ then we see that the integrands are eventually dominated by $(1+x^2)^{-1}$ as $1/n + nx^2 \geq 1 + x^2 \iff x^2 > 1/n$ which is always true in our case for large enough n allowing us to apply DCT and conclude that the limit is zero as the integrands converge to 0 pointwise. However, if $a \leq 0$ then the integrands converge to $\infty \cdot \chi_{\{0\}}$ pointwise which integrates to 0 whereas the limit is that of the terms $\pi/2 - \arctan(na)$ which has a strictly positive limit here for the $a \leq 0$ we consider. This shows us that DCT *can't be applied* here, in fact it is easily proven without the DCT that any function which dominates this sequence can not be L^1 by considering the values on intervals of form $[1/n, 1/(n-1)]$ which is left to the reader as an exercise.

□

Problem 1.29. Show that $\int_0^\infty x^n e^{-x}\,dx = n!$ by differentiating the equation $\int_0^\infty e^{-tx}\,dx = 1/t$. Similarly, show that $\int_{-\infty}^\infty x^{2n}e^{-x^2}\,dx = (2n)!\sqrt{\pi}/4^n n!$ by differentiating the equation $\int_{-\infty}^\infty e^{-tx^2}\,dx = \sqrt{\pi/t}$ (see Proposition 2.53).

Solution. Let $f : [1,\infty) \times [0,\infty) \to \mathbb{R}$ be defined as $f(t,x) = e^{-tx}$, we see that for all t, x we have

$$\left|\frac{\partial}{\partial t} f(x,t)\right| = xe^{-tx} \leq xe^{-x}$$

which is L^1 allowing us to use Theorem 2.27 (page 56) to deduce that

$$\frac{-1}{t^2} = \frac{d}{dt}\int_0^\infty e^{-tx}\mathrm{d}x = \int_0^\infty \frac{\partial}{\partial t} f(x,t)dx = \int_0^\infty -xe^{-tx}dx$$

repeating this a total of n times with the dominating functions being $x^n e^{-x}$ we get the desired conclusion after setting $t = 1$ at the end. For the second parts we work with t over the same interval but this time x varies over $\mathbb{R}$ with $f(t,x) = e^{-tx^2}$ and following the exact same steps with dominating functions of form $x^{2n}e^{-x^2}$ we arrive at a proof. □

Problem 1.30. Show that $\lim_{k\to\infty} \int_0^k x^n(1-k^{-1}x)^k\,dx = n!$.

Solution. The integrands $\chi_{[0,k]}x^n(1-x/k)^k$ are dominated by $x^n e^{-x}$ which is L^1 and also the pointwise limit of the integrands. Applying DCT and using the previosu problem we arrive at the required conclusion. □

Problem 1.31. Derive the following formulas by expanding part of the integrand into an infinite series and justifying the term-by-term integration. Exercise 29 may be useful. (Note: In (d) and (e), term-by-term integration works, and the resulting series converges, only for $a > 1$, but the formulas as stated are actually valid for all $a > 0$.)

a. For $a > 0$, $\int_0^\infty e^{-x^2}\cos ax\,dx = \sqrt{\pi}e^{-a^2/4}$.

b. For $a > -1$, $\int_0^1 x^a(1-x)^{-1}\log x\,dx = \sum_1^\infty (a+k)^{-2}$.

c. For $a > 1$, $\int_0^\infty x^{a-1}(e^x-1)^{-1}\,dx = \Gamma(a)\zeta(a)$, where $\zeta(a) = \sum_1^\infty n^{-a}$.

d. For $a > 1$, $\int_0^\infty e^{-ax} x^{-1} \sin x \, dx = \arctan(a^{-1})$.

e. For $a > 1$, $\int_0^\infty e^{-ax} J_0(x) \, dx = (a^2+1)^{-1/2}$, where $J_0(x) = \sum_0^\infty (-1)^n x^{2n}/4^n (n!)^2$ is the Bessel function of order zero.

Solution. All of these are very easily proven by expanding some function into its power series representation and then using DCT (with the dominating function being of form $\sum |f_n|$ for functions of form $\sum f_n$) to switch the summation and integration so we state what we expand for each.

(a) expand the cosine term and use the previous problem (we set $t = 1$ at the end, here we require the result without that substitution.)

(b) expand the denominator.

(c) Rewrite the integrand as

$$\frac{x^{a-1}}{e^x - 1} = \frac{x^{a-1} e^{-x}}{1 - e^{-x}} = \sum_{n=0}^{\infty} x^{a-1} e^{-x} (e^{-x})^n$$

(d) expand the sine term and use the previous problem.

(e) this is already given as a power series

□

Problem 1.32. Suppose $\mu(X) < \infty$. If f and g are complex-valued measurable functions on X, define

$$\rho(f, g) = \int \frac{|f - g|}{1 + |f - g|} \, d\mu.$$

Then ρ is a metric on the space of measurable functions if we identify functions that are equal a.e., and $f_n \to f$ with respect to this metric iff $f_n \to f$ in measure.

Solution. For the if part, first notice that $x \mapsto x/(1+x)$ is an increasing map on the nonnegatives and thus we have for any arbitrary $\epsilon > 0$ that,

$$\int_X \frac{|f - f_n|}{1 + |f - f_n|} d\mu = \int_{|f - f_n| \geq \epsilon} \frac{|f - f_n|}{1 + |f - f_n|} d\mu + \int_{|f - f_n| < \epsilon} \frac{|f - f_n|}{1 + |f - f_n|} d\mu$$

$$\leq 1 \cdot \mu(\{|f_n - f| \geq \epsilon\}) + \frac{\epsilon}{1 + \epsilon} \cdot \mu(\{|f_n - f| < \epsilon\}) \leq \mu(\{|f_n - f| \geq \epsilon\}) + \epsilon \cdot \mu(X)$$

And as $f_n \to f$ in measure we see that $\lim \rho(f_n, f) \leq \epsilon \mu(X)$ which can be made arbitrarily small proving this direction. For the other direction we similarly have that

$$\rho(f_n, f) \geq \frac{\epsilon}{1 + \epsilon} \cdot \mu(\{|f_n - f| \geq \epsilon\})$$

As $\rho(f_n, f) \to 0$ we must have the right side of the inequality tend to 0 proving the claim. □

Problem 1.33. If $f_n \geq 0$ and $f_n \to f$ in measure, then $\int f \leq \liminf \int f_n$.

Solution. By the definition of the limit infimum we can find a subsequence f_{n_k} such that $\liminf \int f_n = \lim \int f_{n_k}$. As f_n converges in measure to f so does f_{n_k} and we can find a subsequence $f_{n_{k_m}}$ that converges a.e. to f. Now, we see that

$$\liminf_n \int f_n = \lim_k \int f_{n_k} \limsup_k \int f_{n_k} \geq \limsup_m \int f_{n_{k_m}} \geq \liminf_m \int f_{n_{k_m}} \geq \int \liminf f_{n_{k_m}}$$

where we use fatou's lemma for the last inequality. As $f_{n_{k_m}} \to f$ a.e., the last expression is just $\int f$ so we are done. □

Problem 1.34. Suppose $|f_n| \leq g \in L^1$ and $f_n \to f$ in measure.

a. $\int f = \lim \int f_n$.

b. $f_n \to f$ in L^1.

Solution.

(a) By Problem 1.33 we can use fatou's lemma just like we did in the proof for DCT here as well by considering the sequences $g - f_n, g + f_n$ whice are nonnegative and converge to $g - f, g + f$ respectively in measure.

(b) By part (a) and the assumption we clearly see that f_n, f are L^1 and as $|f_n| \to |f|$ in measure too while being dominated by g we can apply part (a) again to see that $\int |f| = \lim \int |f_n|$ which is equivalent to the claim by Problem 1.21.

enumerate □

Problem 1.35. $f_n \to f$ in measure iff for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $\mu(\{x : |f_n(x) - f(x)| \geq \epsilon\}) < \epsilon$ for $n \geq N$.

Solution. The only if direction follows directly from the definition and for the if direction, given $\epsilon > 0$ and $\delta > 0$ let, $\nu = \min\{\epsilon, \delta\}$ we see that $\{|f_n - f| \geq \epsilon\} \subseteq \{|f_n - f| \geq \nu\}$ and we can find some $N \in \mathbb{N}$ such that for all $n \geq N$,

$$\mu(\{|f_n - f| \geq \epsilon\}) \leq \mu(\{|f_n - f| \geq \nu\}) < \nu \leq \delta$$

and we are done. □

Problem 1.36. If $\mu(E_n) < \infty$ for $n \in \mathbb{N}$ and $\chi_{E_n} \to f$ in L^1, then f is (a.e. equal to) the characteristic function of a measurable set.

Problem 1.37. If $f_n \to f$ in measure and $g_n \to g$ in measure, then $f_n + g_n \to f + g$ in measure, and $f_n g_n \to fg$ in measure if $\mu(X) < \infty$, but not necessarily if $\mu(X) = \infty$.

Problem 1.38. If $f_n \to f$ a.e. and $\mu(X) < \infty$, then $f_n \to f$ in measure.

Problem 1.39. If $f_n \to f$ almost uniformly, then $f_n \to f$ a.e. and $f_n \to f$ in measure.

Problem 1.40. In Egoroff's theorem, the hypothesis "$\mu(X) < \infty$" can be replaced by "$|f_n| \leq g$ for all n, where $g \in L^1(\mu)$."

Problem 1.41. If μ is σ-finite and $f_n \to f$ a.e., there exist measurable $E_1, E_2, \cdots \subset X$ such that $\mu((\bigcup_1^\infty E_j)^c) = 0$ and $f_n \to f$ uniformly on each E_j.

Problem 1.42. Let μ be counting measure on $\mathbb{N}$. Then $f_n \to f$ in measure iff $f_n \to f$ uniformly.

Problem 1.43. Suppose that $\mu(X) < \infty$ and $f : X \times [0,1] \to \mathbb{C}$ is a function such that $f(\cdot, y)$ is measurable for each $y \in [0,1]$ and $f(x, \cdot)$ is continuous for each $x \in X$.

a. If $0 < \epsilon, \delta < 1$ then $E_{\epsilon,\delta} = \{x : |f(x,y) - f(x,0)| \leq \epsilon \text{ for all } y < \delta\}$ is measurable.

b. For any $\epsilon > 0$ there is a set $E \subset X$ such that $\mu(E) < \epsilon$ and $f(\cdot, y) \to f(\cdot, 0)$ uniformly on E^c as $y \to 0$.

Problem 1.44. (Lusin's Theorem) If $f : [a,b] \to \mathbb{C}$ is Lebesgue measurable and $\epsilon > 0$, there is a compact set $E \subset [a,b]$ such that $m(E^c) < \epsilon$ and $f|_E$ is continuous. (Use Egoroff's theorem and Theorem 2.26.)

Problem 1.45. If $(X_j, \mathcal{M}_j)$ is a measurable space for $j = 1,2,3$, then $\bigotimes_1^3 \mathcal{M}_j = (\mathcal{M}_1 \otimes \mathcal{M}_2) \otimes \mathcal{M}_3$. Moreover, if μ_j is a σ-finite measure on $(X_j, \mathcal{M}_j)$, then $\mu_1 \times \mu_2 \times \mu_3 = (\mu_1 \times \mu_2) \times \mu_3$.

Problem 1.46. Let $X = Y = [0,1]$, $\mathcal{M} = \mathcal{N} = \mathcal{B}_{[0,1]}$, $\mu =$ Lebesgue measure, and $\nu =$ counting measure. If $D = \{(x,x) : x \in [0,1]\}$ is the diagonal in $X \times Y$, then $\iint \chi_D \, d\mu \, d\nu$, $\iint \chi_D \, d\nu \, d\mu$, and $\int \chi_D \, d(\mu \times \nu)$ are all unequal. (To compute $\int \chi_D \, d(\mu \times \nu)$, go back to the definition of $\mu \times \nu$.)

Problem 1.47. Let $X = Y$ be an uncountable linearly ordered set such that for each $x \in X$, $\{y \in X : y < x\}$ is countable. (Example: the set of countable ordinals.) Let $\mathcal{M} = \mathcal{N}$ be the σ-algebra of countable or co-countable sets, and let $\mu = \nu$ be defined on $\mathcal{M}$ by $\mu(A) = 0$ if A is countable and $\mu(A) = 1$ if A is co-countable. Let $E = \{(x,y) \in X \times X : y < x\}$. Then E_x and E^y are measurable for all x, y, but $\iint \chi_E \, d\mu \, d\nu$ and $\iint \chi_E \, d\nu \, d\mu$ exist but are not equal. (If one believes in the continuum hypothesis, one can take $X = [0,1]$ [with a nonstandard ordering] and thus obtain a set $E \subset [0,1]^2$ such that E_x is countable and E^y is co-countable [in particular, Borel] for all x, y, but E is not Lebesgue measurable.)

Problem 1.48. Let $X = Y = \mathbb{N}, \mathcal{M} = \mathcal{N} = \mathcal{P}(\mathbb{N}), \mu = \nu =$ counting measure. Define $f(m,n) = 1$ if $m = n, f(m,n) = -1$ if $m = n+1$, and $f(m,n) = 0$ otherwise. Then $\int f \, d\mu \, d\nu$ and $\int f \, d\nu \, d\mu$ exist and are unequal.

Problem 1.49. Prove Theorem 2.39 by using Theorem 2.37 and Proposition 2.12 together with the following lemmas.

a. If $E \in \mathcal{M} \times \mathcal{N}$ and $\mu \times \nu(E) = 0$, then $\nu(E_x) = 0$ for a.e. x and $\mu(E^y) = 0$ for a.e. y.

b. If f is $\mathcal{M}$-measurable and $f = 0$ λ-a.e., then f_x and f^y are integrable for a.e. x and y, and $\int f_x \, d\nu = \int f^y \, d\mu = 0$ for a.e. x and y. (Here the completeness of μ and ν is needed.)

Problem 1.50. Suppose $(X, \mathcal{M}, \mu)$ is a σ-finite measure space and $f \in L^+(X)$. Let

$$G_f = \{(x,y) \in X \times [0,\infty] : y \leq f(x)\}.$$

Then G_f is $\mathcal{M} \times \mathcal{B}_{\mathbb{R}}$-measurable and $\mu \times m(G_f) = \int f\,d\mu$; the same is also true if the inequality $y \le f(x)$ in the definition of G_f is replaced by $y < f(x)$. (To show measurability of G_f, note that the map $(x,y) \mapsto f(x) - y$ is the composition of $(x,y) \mapsto (f(x),y)$ and $(z,y) \mapsto z - y$.) This is the definitive statement of the familiar theorem from calculus, "the integral of a function is the area under its graph."

Problem 1.51. Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be arbitrary measure spaces (not necessarily σ-finite).

a. If $f : X \to \mathbb{C}$ is $\mathcal{M}$-measurable, $g : Y \to \mathbb{C}$ is $\mathcal{N}$-measurable, and $h(x,y) = f(x)g(y)$, then h is $\mathcal{M} \otimes \mathcal{N}$-measurable.

b. If $f \in L^1(\mu)$ and $g \in L^1(\nu)$, then $h \in L^1(\mu \times \nu)$ and $\int h\,d(\mu \times \nu) = (\int f\,d\mu)(\int g\,d\nu)$.

Problem 1.52. The Fubini-Tonelli theorem is valid when $(X, \mathcal{M}, \mu)$ is an arbitrary measure space and Y is a countable set, $\mathcal{N} = \mathcal{P}(Y)$, and ν is counting measure on Y. (Cf. Theorems 2.15 and 2.25.)

Problem 1.53. Fill in the details of the proof of Theorem 2.41.

Problem 1.54. How much of Theorem 2.44 remains valid if T is not invertible?

Problem 1.55. Let $E = [0,1] \times [0,1]$. Investigate the existence and equality of $\int_E f\,dm^2$, $\int_0^1 \int_0^1 f(x,y)\,dx\,dy$, and $\int_0^1 \int_0^1 f(x,y)\,dy\,dx$ for the following f:

a. $f(x,y) = (x^2 - y^2)(x^2 + y^2)^{-2}$.

b. $f(x,y) = (1 - xy)^{-a}$ $(a > 0)$.

c. $f(x,y) = (x - \frac{1}{2})^{-3}$ if $0 < y < |x - \frac{1}{2}|$, $f(x,y) = 0$ otherwise.

Problem 1.56. If f is Lebesgue integrable on $(0,a)$ and $g(x) = \int_x^a t^{-1} f(t)\,dt$, then g is integrable on $(0,a)$ and $\int_0^a g(x)\,dx = \int_0^a f(x)\,dx$.

Problem 1.57. Show that $\int_0^\infty e^{-sx} x^{-1} \sin x\,dx = \arctan(s^{-1})$ for $s > 0$ by integrating $e^{-sxy} \sin x$ with respect to x and y. (It may be useful to recall that $\tan\left(\frac{\pi}{2} - \theta\right) = (\tan\theta)^{-1}$. Cf. Exercise 31d.)

Problem 1.58. Show that $\int e^{-sx} x^{-1} \sin^2 x\,dx = \frac{1}{4}\log(1 + 4s^{-2})$ for $s > 0$ by integrating $e^{-sx} \sin 2xy$ with respect to x and y.

Problem 1.59. Let $f(x) = x^{-1} \sin x$.

a. Show that $\int_0^\infty |f(x)|\,dx = \infty$.

b. Show that $\lim_{b\to\infty} \int_0^b f(x)\,dx = \frac{1}{2}\pi$ by integrating $e^{-xy} \sin x$ with respect to x and y. (In view of part (a), some care is needed in passing to the limit as $b \to \infty$.)

Problem 1.60. $\Gamma(x)\Gamma(y)/\Gamma(x+y) = \int_0^1 t^{x-1}(1-t)^{y-1}\,dt$ for $x, y > 0$. (Recall that Γ was defined in §2.3. Write $\Gamma(x)\Gamma(y)$ as a double integral and use the argument of the exponential as a new variable of integration.)

Problem 1.61. If f is continuous on $[0,\infty)$, for $\alpha > 0$ and $x \geq 0$ let

$$I_\alpha f(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t)\, dt.$$

$I_\alpha f$ is called the αth **fractional integral** of f.

a. $I_{\alpha+\beta} f = I_\alpha(I_\beta f)$ for all $\alpha, \beta > 0$. (Use Exercise 60.)

b. If $n \in \mathbb{N}$, $I_n f$ is an nth-order antiderivative of f.

Problem 1.62. The measure σ on S^{n-1} is invariant under rotations.

Problem 1.63. The technique used to prove Proposition 2.54 can also be used to integrate any polynomial over S^{n-1}. In fact, suppose $f(x) = \prod_1^n x_j^{\alpha_j}$ ($\alpha_j \in \mathbb{N} \cup \{0\}$) is a monomial. Then $\int f\, d\sigma = 0$ if any α_j is odd, and if all α_j's are even,

$$\int f\, d\sigma = \frac{2\Gamma(\beta_1)\dots\Gamma(\beta_n)}{\Gamma(\beta_1 + \dots + \beta_n)} \quad \text{where } \beta_j = \frac{\alpha_j + 1}{2}.$$

Problem 1.64. For which real values of a and b is $|x|^a |\log|x||^b$ integrable over $\{x \in \mathbb{R}^n : |x| < \frac{1}{2}\}$? Over $\{x \in \mathbb{R}^n : |x| > 2\}$?

Problem 1.65. Define $G : \mathbb{R}^n \to \mathbb{R}^n$ by $G(r, \phi_1, \dots, \phi_{n-2}, \theta) = (x_1, \dots, x_n)$ where

$$\begin{aligned} x_1 &= r\cos\phi_1, \quad x_2 = r\sin\phi_1\cos\phi_2, \quad x_3 = r\sin\phi_1\sin\phi_2\cos\phi_3, \dots, \\ x_{n-1} &= r\sin\phi_1\dots\sin\phi_{n-2}\cos\theta, \quad x_n = r\sin\phi_1\dots\sin\phi_{n-2}\sin\theta. \end{aligned}$$

a. G maps $\mathbb{R}^n$ onto $\mathbb{R}^n$, and $|G(r, \phi_1, \dots, \theta)| = |r|$.

b. $\det D_G(r, \phi_1, \dots, \theta) = r^{n-1} \sin^{n-2}\phi_1 \sin^{n-3}\phi_2 \dots \sin\phi_{n-2}$.

c. Let $\Omega = (0,\infty) \times (0,\pi)^{n-2} \times (0, 2\pi)$. Then $G|_\Omega$ is a diffeomorphism and $m(\mathbb{R}^n \setminus G(\Omega)) = 0$.

d. Let $F(\phi_1, \dots, \phi_{n-2}, \theta) = G(1, \phi_1, \dots, \phi_{n-2}, \theta)$ and $\Omega' = (0,\pi)^{n-2} \times (0, 2\pi)$. Then $(F|\Omega')^{-1}$ defines a coordinate system on S^{n-1} except on a σ-null set, and the measure σ is given in these coordinates by

$$d\sigma(\phi_1, \dots, \phi_{n-2}, \theta) = \sin^{n-2}\phi_1 \sin^{n-3}\phi_2 \dots \sin\phi_{n-2}\, d\phi_1 \dots d\phi_{n-2}\, d\theta.$$